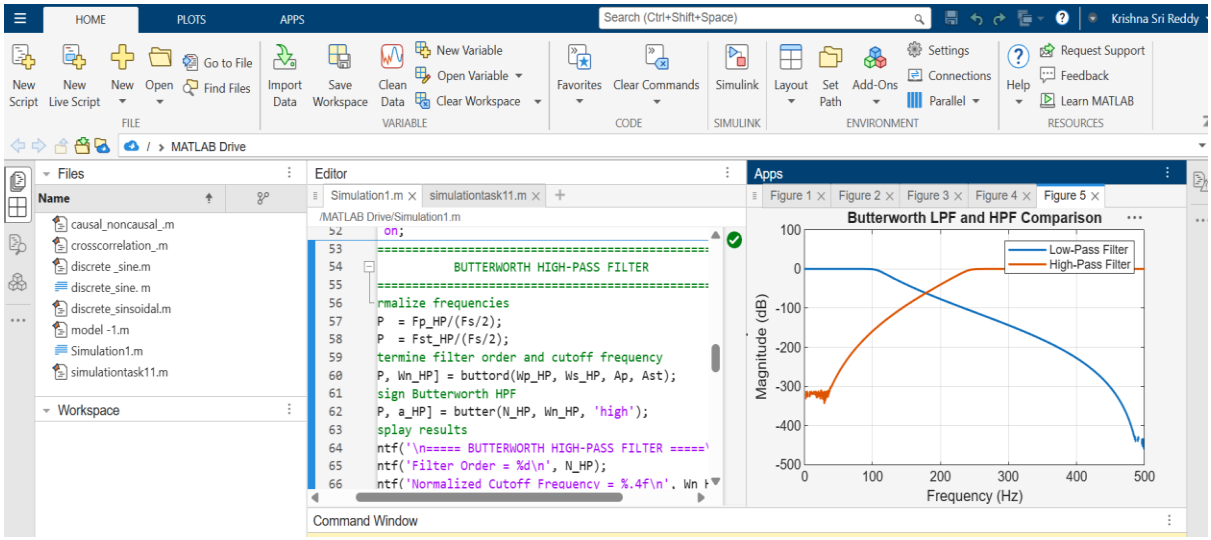
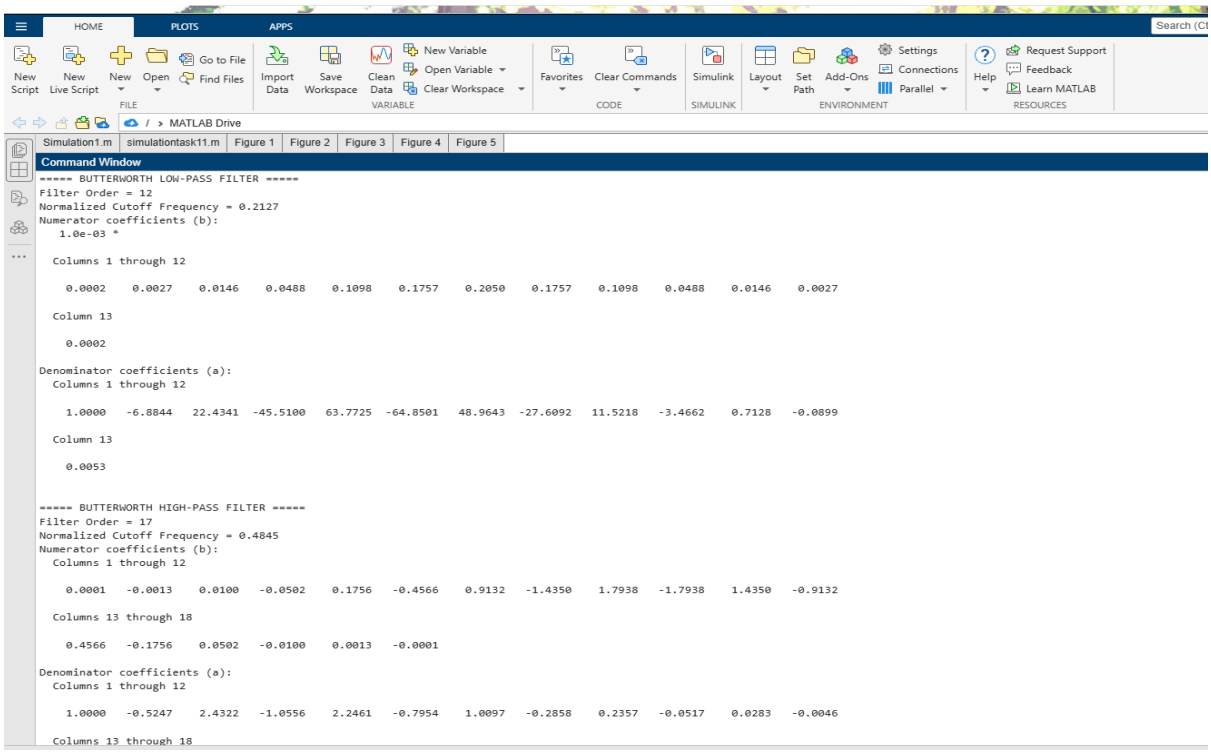

HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
Krishna Sri Reddy
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
Add-Ons
Settings
Connections
Parallel
Help
Request Support
Feedback
Learn MATLAB
FILE
VARIABLE
CODE
SIMULINK
ENVIRONMENT
RESOURCES
/ > MATLAB Drive
Files
Name
causal_noncausal_m
crosscorrelation_m
discrete _sine.m
discrete_sine. m
discrete_sinsoidal.m
model -1.m
Simulation1.m
simulationtask11.m
Workspace
Editor
Simulation1.m
simulationtask11.m
/MATLAB Drive/Simulation1.m
on;
BUTTERWORTH HIGH-PASS FILTER
rmalize frequencies
P = Fp_HP/(Fs/2);
P = Fst_HP/(Fs/2);
termine filter order and cutoff frequency
P, Wn_HP] = buttord(Wp_HP, Ws_HP, Ap, Ast);
sign Butterworth HPF
P, a_HP] = butter(N_HP, Wn_HP, 'high');
splay results
ntf('\n===== BUTTERWORTH HIGH-PASS FILTER =====\
ntf('Filter Order = %d\n', N_HP);
ntf('Normalized Cutoff Frequency = %.4f\n', Wn
Command Window
Apps
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Butterworth LPF and HPF Comparison
Low-Pass Filter
High-Pass Filter
Magnitude (dB)
Frequency (Hz)
100
0
-100
-200
-300
-400
-500
0
100
200
300
400
500


HOME
PLOTS
APPS
Search (Ct
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
Add-Ons
Settings
Connections
Parallel
Help
Request Support
Feedback
Learn MATLAB
FILE
VARIABLE
CODE
SIMULINK
ENVIRONMENT
RESOURCES
/ > MATLAB Drive
Simulation1.m
simulationtask11.m
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Command Window
===== BUTTERWORTH LOW-PASS FILTER =====
Filter Order = 12
Normalized Cutoff Frequency = 0.2127
Numerator coefficients (b):
1.0e-03 *
Columns 1 through 12
0.0002 0.0027 0.0146 0.0488 0.1098 0.1757 0.2050 0.1757 0.1098 0.0488 0.0146 0.0027
Column 13
0.0002
Denominator coefficients (a):
Columns 1 through 12
1.0000 -6.8844 22.4341 -45.5100 63.7725 -64.8501 48.9643 -27.6092 11.5218 -3.4662 0.7128 -0.0899
Column 13
0.0053
===== BUTTERWORTH HIGH-PASS FILTER =====
Filter Order = 17
Normalized Cutoff Frequency = 0.4845
Numerator coefficients (b):
Columns 1 through 12
0.0001 -0.0013 0.0100 -0.0502 0.1756 -0.4566 0.9132 -1.4350 1.7938 -1.7938 1.4350 -0.9132
Columns 13 through 18
0.4566 -0.1756 0.0502 -0.0100 0.0013 -0.0001
Denominator coefficients (a):
Columns 1 through 12
1.0000 -0.5247 2.4322 -1.0556 2.2461 -0.7954 1.0097 -0.2858 0.2357 -0.0517 0.0283 -0.0046
Columns 13 through 18